

Notes: He & Tian (JFE, 2013)

The Dark Side of Analyst Coverage: The Case of Innovation

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1 Big picture

- **Question.** Do financial analysts help or hurt corporate innovation?
- **Tension.** Analysts reduce information asymmetry, which could make innovative firms easier to value and finance. But analysts also impose short-term earnings pressure, which can push managers away from risky, long-horizon innovation projects.
- **Main finding.** Firms followed by more analysts generate fewer patents and lower-impact patents. The negative relation remains after firm fixed effects, year fixed effects, and a rich set of controls.
- **Causal evidence.** The paper uses exogenous reductions in analyst coverage caused by brokerage closures and brokerage mergers. When analyst coverage falls, treatment firms increase innovation relative to matched control firms.
- **IV evidence.** The paper also instruments analyst coverage using expected coverage driven by brokerage-house size changes. Two-stage least squares estimates are negative and larger than OLS estimates.
- **Mechanism.** Analysts appear to create managerial myopia. They impose external pressure to meet short-term earnings expectations, making long-run, risky innovation less attractive.
- **Additional mechanisms.** The paper also examines institutional ownership, takeover exposure, stock illiquidity, and discretionary accruals. These channels explain part of the effect, but analyst coverage has a residual negative effect even after controlling for them.
- **Economic takeaway.** Financial analysts are not only information intermediaries. They can have real effects on firms' investment choices, and their short-term monitoring can harm innovation.

2 Incremental contribution of the paper

2.1 Contribution relative to the analyst literature

Most research on financial analysts emphasizes their capital-market roles: information production, stock recommendations, forecast accuracy, liquidity, and price discovery. This paper adds a real-effects dimension. It shows that analysts affect not only how investors value firms but also what firms choose to do. Specifically, analyst coverage changes firms' incentives to invest in long-run innovation.

Interpretation: The incremental contribution is to move analyst research from the information environment to corporate policy. Analysts can improve informational efficiency while simultaneously distorting real investment.

2.2 Contribution relative to the innovation literature

The innovation literature studies financing constraints, institutional ownership, product market competition, legal protection, takeovers, and private versus public equity. This paper adds analyst coverage as a distinct public-market force that affects innovation.

- Innovation is hard to evaluate, long-horizon, risky, and often not immediately reflected in earnings.
- Analysts focus heavily on near-term earnings forecasts and stock recommendations.
- This creates a conflict between the innovation horizon and the analyst-evaluation horizon.

Interpretation: The paper's contribution is not merely to add another control variable. It introduces analyst pressure as a mechanism that links public-market scrutiny to technological output.

2.3 Contribution relative to managerial myopia research

Prior work documents that managers may sacrifice long-run value to meet short-term targets. He and Tian connect that idea to an externally observable market institution: analyst coverage. They show that the pressure associated with analyst following can reduce patent quantity and quality.

Interpretation: This is a concrete empirical setting where short-term market pressure has measurable effects on long-run investment output.

2.4 Contribution relative to identification in corporate finance

The paper uses two complementary identification strategies:

1. A difference-in-differences design based on brokerage closures and mergers.
2. An instrumental-variable design based on expected analyst coverage from broker-size changes.

Interpretation: The incremental identification contribution is important because the raw relation between coverage and innovation is endogenous. Innovative firms may attract more analysts, and unobserved firm quality can affect both variables. The paper’s designs address this concern from two separate angles.

2.5 Contribution relative to the “dark side of finance” literature

The paper contributes to the idea that financial development can have adverse externalities. Analysts reduce information frictions, but the same institution can also intensify short-termism.

Interpretation: The paper’s broader contribution is to show that financial intermediation is not uniformly beneficial for real investment. The effect depends on which activities financial intermediaries emphasize and how managers respond.

3 Data and measurement

3.1 Sample construction

- The sample consists of U.S. public firms from 1993 to 2005.
- Patent and citation data come from patent databases used in the innovation literature.
- Analyst coverage comes from I/B/E/S and is based on the number of analysts issuing earnings forecasts.
- Accounting variables come from Compustat; stock market variables come from CRSP; institutional ownership comes from 13F filings.
- The baseline sample has 25,860 firm-year observations.

Interpretation: The sample period is chosen because it provides overlapping firm-year data on analyst coverage, patents, citations, and firm controls. The outcome variables are measured three years ahead because innovation is a long-term process.

3.2 Innovation outcomes

The two main innovation outcomes are:

$$\ln Patent_{i,t+3} = \ln(1 + Patent_{i,t+3}), \quad (1)$$

where $Patent_{i,t+3}$ is the total number of patents filed by firm i in year $t + 3$ and eventually granted.

The second outcome is:

$$LnCitePat_{i,t+3} = \ln(1 + CitePat_{i,t+3}), \quad (2)$$

where *CitePat* is the number of non-self citations received by a firm's patents, scaled by the number of patents.

Interpretation: Patents proxy for innovation quantity. Citations per patent proxy for innovation quality or impact. The three-year-ahead measurement helps capture the long horizon of innovation projects.

3.3 Analyst coverage

Analyst coverage is defined as the average monthly number of earnings forecasts issued for firm *i* during fiscal year *t*:

$$Coverage_{i,t} = \frac{1}{12} \sum_{m=1}^{12} Analysts_{i,m,t}. \quad (3)$$

The main regressor is:

$$LnCoverage_{i,t} = \ln(1 + Coverage_{i,t}). \quad (4)$$

Interpretation: The logarithmic transformation dampens the effect of extreme analyst-following firms and allows the marginal effect of another analyst to decline with the initial level of coverage.

3.4 Baseline controls

The paper controls for firm characteristics that may affect innovation:

- firm size, measured by assets;
- R&D intensity;
- firm age;
- profitability, measured by ROA;
- tangible assets and capital expenditures;
- leverage;
- Tobin's *Q*;
- financial constraints, measured by the Kaplan–Zingales index;
- industry concentration, measured by the Herfindahl index;
- institutional ownership;
- stock illiquidity.

Interpretation: These controls address standard determinants of innovation: resources, investment opportunities, financial constraints, industry structure, and governance.

4 Empirical specifications and identification

4.1 Baseline firm-panel specification

The baseline regression is:

$$Innovation_{i,t+3} = \alpha_i + \delta_t + \beta LnCoverage_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t}, \quad (5)$$

where *Innovation* is either *LnPatent* or *LnCitePat*, α_i are firm fixed effects, and δ_t are year fixed effects.

Interpretation: Firm fixed effects compare the same firm over time. Year fixed effects remove aggregate innovation trends. A negative β means that increases in analyst coverage within a firm are associated with lower future innovation.

4.2 Difference-in-differences using brokerage closures and mergers

The first causal design exploits exogenous shocks to coverage from brokerage closures and mergers.

- Brokerage closures reduce coverage when a firm was followed by analysts employed by a closing broker.
- Brokerage mergers reduce coverage when analysts become redundant after the merger.
- These shocks are plausibly unrelated to the innovation opportunities of the covered firms.

For each treated firm, the paper constructs matched controls using market capitalization, book-to-market, past return, analyst coverage, stock return volatility, and turnover.

The DiD estimator compares innovation before and after the event:

$$DiD = (Innovation_{post}^T - Innovation_{pre}^T) - (Innovation_{post}^C - Innovation_{pre}^C). \quad (6)$$

Interpretation: A positive DiD estimate means that firms losing analyst coverage increase innovation relative to matched controls. Since the treatment is a reduction in coverage, a positive DiD implies a negative causal effect of analyst coverage on innovation.

4.3 Parallel trends diagnostics

The paper checks whether treatment and control firms have similar pre-event innovation trends. It plots treatment-minus-control innovation outcomes from year -3 to year $+3$.

Interpretation: The identifying assumption is that, absent the analyst-coverage shock, treatment and control firms would have followed similar innovation trajectories. The pre-event trends are stable, supporting the design.

4.4 Heterogeneity by initial analyst coverage

The paper divides treated firms by initial analyst coverage:

- firms with at most ten analysts;
- firms with 10–25 analysts;
- firms with more than 25 analysts.

Interpretation: The effect is strongest when firms initially have relatively low or moderate coverage. When many analysts already cover the firm, losing one analyst has less impact.

4.5 Instrumental-variable strategy

The second causal design instruments analyst coverage with expected coverage, following the idea that changes in brokerage-house size predict changes in analyst coverage.

Expected coverage is constructed as:

$$ExpCoverage_{i,t} = \sum_j Coverage_{i,j,0} \times \frac{BrokerSize_{j,t}}{BrokerSize_{j,0}}, \quad (7)$$

where $Coverage_{i,j,0}$ indicates broker j 's initial coverage of firm i , and $BrokerSize_{j,t}$ is the number of analysts employed by broker j in year t .

The first stage is:

$$LnCoverage_{i,t} = \pi_0 + \pi_1 ExpCoverage_{i,t} + \Pi' X_{i,t} + \alpha_i + \delta_t + u_{i,t}. \quad (8)$$

The second stage is:

$$Innovation_{i,t+3} = \beta \widehat{LnCoverage}_{i,t} + \gamma' X_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}. \quad (9)$$

Interpretation: The instrument varies with brokerage-house expansion or contraction rather than firm-specific innovation opportunities. The exclusion restriction is that broker-size changes affect a firm's innovation only through analyst coverage.

5 Mechanisms

5.1 Dedicated versus short-term institutional ownership

The paper argues that a decline in analyst coverage can change the composition of institutional ownership.

- Dedicated institutional investors may have private information and longer horizons, so they can shield managers from short-term pressure.
- Transient or quasi-indexing investors are less likely to monitor long-term innovation and may chase short-term performance.
- Analyst coverage can crowd out private information acquisition by dedicated investors.

Interpretation: If losing analyst coverage attracts dedicated investors and reduces short-term investors, managers may gain more tolerance for long-term innovation.

5.2 Takeover exposure

Analyst coverage can affect takeover exposure because analysts make firms more visible and easier to value. The paper tests whether reductions in analyst coverage lower takeover pressure.

Interpretation: Less takeover pressure can encourage managers to invest in projects with long horizons and uncertain payoff, such as innovation.

5.3 Stock liquidity

Analyst coverage increases information production and can improve liquidity. More liquidity may make informed trading easier and increase pressure on managers through the stock market.

Interpretation: If lower coverage reduces liquidity, managers may face less short-term market discipline and become more willing to invest in innovation.

5.4 Discretionary accruals and earnings management

The paper also tests whether analyst coverage affects innovation through accrual-based earnings management. Discretionary accruals are measured using a Yu-style accrual model.

Interpretation: If analysts pressure managers to meet earnings targets, managers may manipulate accruals or cut innovation. The paper asks whether innovation effects remain after controlling for accrual management.

5.5 Residual effect after mechanism controls

The paper finds that the treatment effect remains positive after controlling for institutional ownership changes, takeover exposure, stock illiquidity, and discretionary accruals.

Interpretation: These mechanisms account for part of the effect but not all of it. There is a residual direct effect of analyst coverage on innovation consistent with short-term pressure.

6 Tables and figures

6.1 Table 1: Variable definitions and summary statistics

Read:

- The baseline sample contains 25,860 firm-year observations.
- Average annual patent count is 9.785, while the median is zero, showing strong skewness in innovation output.
- Average analyst coverage is 7.041 analysts; the median is 4.583.
- The table defines the key outcome variables, controls, and mechanism variables used throughout the paper.

Economic message: The summary statistics show that innovation is concentrated in a subset of firms and that analyst coverage varies substantially across public firms.

Panel A: Variable definition						
Variable	Definition					
Measures of innovation						
$\ln Patent_{i,t}$	Natural logarithm of one plus firm i 's total number of patents filed (and eventually granted) in year $t+3$					
$\ln CitePat_{i,t}$	Natural logarithm of one plus firm i 's total number of non-self citations received on the firm's patents filed (and eventually granted), scaled by the number of the patents filed (and eventually granted) in year $t+3$					
Measures of analyst coverage and other variables						
$Coverage_{i,t}$	Arithmetic mean of the 12 monthly numbers of earnings forecasts for firm i extracted from the Institutional Brokers' Estimate System summary file over fiscal year t					
$Assets_{i,t}$	Book value of total assets (Compustat data item #6) measured at the end of fiscal year t					
$R\&D/Assets_{i,t}$	Research and development (R&D) expenditure (#46) divided by book value of total assets (#6) measured at the end of fiscal year t , set to 0 if missing					
$Age_{i,t}$	Firm i 's age, approximated by the number of years listed on Compustat					
$ROA_{i,t}$	Return on assets ratio defined as operating income before depreciation (#13) divided by book value of total assets (#6), measured at the end of fiscal year t					
$PPE/Assets_{i,t}$	Property, plant & equipment (#8) divided by book value of total assets (#6) measured at the end of fiscal year t					
$Leverage_{i,t}$	Firm i 's leverage ratio, defined as book value of debt (#9+434) divided by book value of total assets (#6) measured at the end of fiscal year t					
$Capex/Assets_{i,t}$	Capital expenditure (#128) scaled by book value of total assets (#6) measured at the end of fiscal year t					
$TobinQ_{i,t}$	Firm i 's market-to-book ratio during fiscal year t , calculated as market value of equity (#199 + #25) plus book value of assets (#6) minus book value of equity (#60) minus balance sheet deferred taxes (#74, set to zero if missing), divided by book value of assets (#6)					
$KZindex_{i,t}$	Firm i 's Kaplan and Zingales index measured at the end of fiscal year t , calculated as $-1.002 \times \text{cash flow} (\#18 + \#14) / \#6$ plus $0.283 \times Q / (\#5 + \#109 + \#25 - \#60 - \#74) / \#6$ plus $3.139 \times \text{leverage} (\#9 + \#344 + \#216) / \#6$ minus $35.368 \times \text{dividends} (\#21 + \#19) / \#6$ minus $1.315 \times \text{cash holdings} (\#148)$, where #8 is lagged					
$Hindex_{i,t}$	Herfindahl index of four-digit standard industrial classification (SIC) industry i where firm i belongs, measured at the end of fiscal year t					
$InstOwn_{i,t}$	Institutional holdings (percent) for firm i over fiscal year t , calculated as the arithmetic mean of the four quarterly institutional holdings reported through form 13F					
$Illiquidity_{i,t}$	Natural logarithm of relative effective spread measured over firm i 's fiscal year t , where relative effective spread is defined as the absolute value of the difference between the execution price and the mid-point of the prevailing bid-ask quote divided by the midpoint of the prevailing bid-ask quote					
$ExpCoverage_{i,t}$	Natural logarithm of the sum of expected analyst coverage from all brokers covering firm i in year t , where the expected coverage from broker j is the product of the analyst coverage from broker j for firm i in year 0 and the ratio of broker j 's size (total number of analysts employed by the broker) in year t to broker j 's size in year 0					
$InstOwn_{i,t}$	Institutional holdings (percent) for firm i over fiscal year t held by dedicated institutional investors, per Bushee (2001) classification					
$TransOwn_{i,t}$	Institutional holdings (percent) for firm i over fiscal year t held by transient institutional investors and quasi-investors, per Bushee (2001) classification					
$DisAccrual_{i,t}$	Absolute value of the difference between $TA/Assets_{i,t}$ and $NDIA_{i,t}$ is estimated by $TA_{i,t}/Assets_{i,t-1} = \alpha_1 1/Assets_{i,t-1} + \alpha_2 \Delta REV_{i,t}/Assets_{i,t-1} + \alpha_3 PPE_{i,t}/Assets_{i,t-1} + \alpha_4 \Delta R\&D_{i,t}/Assets_{i,t-1} + \alpha_5 \Delta R_{i,t}/Assets_{i,t-1} + \alpha_6 \Delta R_{i,t}/Assets_{i,t-1}$, where ΔR is the changes in receivables (#2)					
Panel B: Summary statistics						
Variable	25th percentile	Median	Mean	75th percentile	Standard deviation	N
Patent	0.000	0.000	9.785	2.652	30.044	25,860
CitePat	0.000	0.000	3.884	3.579	8.375	25,860
Coverage	2.083	4.583	7.041	9.583	6.697	25,860
$\ln Coverage$	0.734	1.522	1.515	2.260	0.964	25,860
Assets	0.111	0.426	3.593	1.859	10.690	25,860
$R\&D/Assets$	0.000	0.000	0.050	0.061	0.096	25,860
Age	6.000	11.000	17.087	27.080	14.701	25,860
ROA	0.066	0.125	0.095	0.182	0.186	25,860
$PPE/Assets$	0.104	0.225	0.297	0.443	0.239	25,860
Leverage	0.030	0.192	0.218	0.346	0.198	25,860
$Capex/Assets$	0.025	0.047	0.063	0.083	0.064	25,860
TobinQ	1.120	1.521	2.141	2.392	1.788	25,860
KZindex	-5.104	-0.891	-6.306	0.771	19.427	25,860
Hindex	0.094	0.236	0.325	0.467	0.282	25,860
InstOwn	0.221	0.443	0.438	0.647	0.255	25,860
Illiquidity	-5.691	-4.850	-4.854	-3.970	1.150	25,860
ExpCoverage	1.386	1.846	1.923	2.580	0.740	13,673
TransOwn	0.015	0.073	0.097	0.150	0.095	25,860
DisAccrual	0.151	0.333	0.345	0.515	0.224	25,860

Table 2
Baseline regression of innovation outcomes on analyst coverage.
This table reports regressions of the innovation outcome variables (three-year-ahead number of patents and number of non-self citations per patent) on analyst coverage and other control variables. Definitions of variables are in Panel A of Table 1. Each regression includes a separate intercept. Robust standard errors clustered by firm are displayed in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Patents					
Dependent variable	$\ln Patent_{i,t+3}$ (1)	$\ln Patent_{i,t+3}$ (2)	$\ln Patent_{i,t+3}$ (3)	$\ln Patent_{i,t+3}$ (4)	$\ln Patent_{i,t+3}$ (5)
$\ln Coverage$	0.371*** (0.025)	-0.026* (0.015)	-0.051*** (0.016)	-0.053*** (0.016)	-0.052*** (0.017)
$\ln Assets$		0.040*** (0.020)	0.050*** (0.020)	0.058*** (0.020)	0.058*** (0.022)
$R\&D/Assets$		0.093* (0.048)	0.100** (0.048)	0.086* (0.048)	0.086* (0.048)
$\ln Age$		0.244*** (0.072)	0.180*** (0.072)	0.180*** (0.072)	0.180*** (0.072)
ROA		0.705*** (0.200)	0.693*** (0.200)	0.719*** (0.200)	0.719*** (0.201)
$PPE/Assets$		0.283*** (0.106)	0.330*** (0.106)	0.327*** (0.105)	0.327*** (0.105)
$Leverage$		-0.336*** (0.067)	-0.324*** (0.067)	-0.331*** (0.067)	-0.331*** (0.067)
$Capex/Assets$		-0.014 (0.114)	-0.014 (0.114)	-0.016 (0.115)	-0.016 (0.115)
$TobinQ$		0.019*** (0.005)	0.020*** (0.005)	0.020*** (0.005)	0.020*** (0.005)
$KZindex$		-0.001** (0.000)	-0.001** (0.000)	-0.001** (0.000)	-0.001** (0.000)
$Hindex$		0.226 (0.163)	0.226 (0.163)	0.226 (0.163)	0.226 (0.163)
$Hindex Squared$		-0.128 (0.139)	-0.128 (0.139)	-0.128 (0.139)	-0.128 (0.139)
$InstOwn$		0.040** (0.020)	0.040** (0.020)	0.040** (0.020)	0.040** (0.020)
$Illiquidity$		0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Firm fixed effects	No	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	25,860	25,860	25,860	25,860	25,860
R^2	0.097	0.832	0.833	0.833	0.833
Panel B: Citations per patent					
Dependent variable	$\ln CitePat_{i,t+3}$ (1)	$\ln CitePat_{i,t+3}$ (2)	$\ln CitePat_{i,t+3}$ (3)	$\ln CitePat_{i,t+3}$ (4)	$\ln CitePat_{i,t+3}$ (5)
$\ln Coverage$	0.124*** (0.011)	-0.070*** (0.018)	-0.083*** (0.018)	-0.080*** (0.018)	-0.077*** (0.019)
$\ln Assets$		-0.011 (0.025)	-0.015 (0.025)	-0.015 (0.025)	-0.015 (0.027)
$R\&D/Assets$		0.221*** (0.061)	0.225*** (0.061)	0.212*** (0.061)	0.212*** (0.061)
$\ln Age$		0.298*** (0.085)	0.260*** (0.085)	0.260*** (0.087)	0.260*** (0.088)
ROA		0.668*** (0.221)	0.765*** (0.221)	0.791*** (0.224)	0.791*** (0.224)
$PPE/Assets$		0.287*** (0.134)	0.318*** (0.133)	0.318*** (0.133)	0.313** (0.133)
$Leverage$		-0.304*** (0.074)	-0.288*** (0.074)	-0.288*** (0.075)	-0.288*** (0.075)
$Capex/Assets$		0.432*** (0.155)	0.426*** (0.155)	0.466*** (0.155)	0.466*** (0.155)
$TobinQ$		-0.000 (0.006)	-0.000 (0.006)	-0.000 (0.006)	-0.000 (0.006)
$KZindex$		-0.001 (0.187)	-0.001 (0.187)	-0.001 (0.187)	-0.001 (0.187)
$Hindex$		-0.117 (0.143)	-0.117 (0.143)	-0.117 (0.143)	-0.117 (0.143)
$Hindex Squared$		-0.118 (0.143)	-0.118 (0.143)	-0.118 (0.143)	-0.118 (0.143)

6.2 Table 2: Baseline relation between analyst coverage and innovation

Read:

- Without firm fixed effects, $\ln Coverage$ is positively related to patents, reflecting cross-sectional selection: innovative firms tend to attract analyst coverage.
- Once firm fixed effects are included, the coefficient turns negative.
- In the full patent specification, $\ln Coverage$ is about -0.052 and statistically significant.
- For citations per patent, the full specification gives a coefficient around -0.077 and statistically significant.
- The sign switch highlights the importance of controlling for time-invariant firm quality.

Economic message: The baseline panel evidence suggests that increases in coverage within a firm are associated with less innovation in future years.

Table 2 (*continued*)

Panel B: Citations per patent					
Dependent variable	$LnCitePat_{t+3}$ (1)	$LnCitePat_{t+3}$ (2)	$LnCitePat_{t+3}$ (3)	$LnCitePat_{t+3}$ (4)	$LnCitePat_{t+3}$ (5)
<i>Illiquidity</i>					(0.087) 0.043** (0.021)
Firm fixed effects	No	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	25,860	25,860	25,860	25,860	25,860
R^2	0.157	0.615	0.618	0.619	0.620

and stock illiquidity are positively related to innovation, corporate innovation could bias our results. To that end, we

6.3 Figure 1 and Table 3: Difference-in-differences evidence

Read:

- Figure 1 shows no obvious pre-trend in treatment-minus-control innovation before the analyst-coverage shock.
- After the shock, treatment firms increase patents and citations relative to controls.
- Table 3 reports a patent DiD estimate of 0.182 and a citation DiD estimate of 0.294, both significant at the 1% level.
- The effect is strongest for firms with ten or fewer analysts and weaker for firms with more than 25 analysts.

Economic message: Losing analyst coverage increases innovation relative to matched firms, supporting a negative causal effect of analyst coverage.

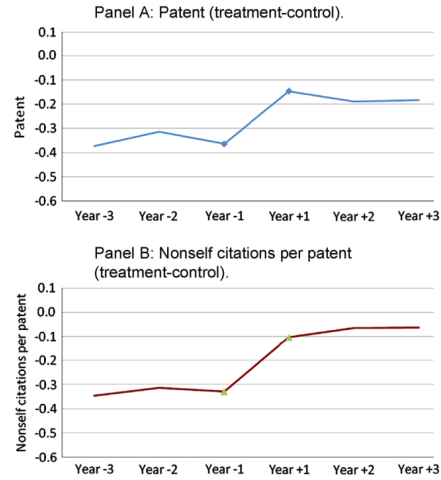


Fig. 1. Trends of innovation activities in the treatment sample (net of control). This figure shows the trend of innovation activities for the treatment sample net of the control group three years before and after the event (broker closure or merger) year. Panel A shows the trend for the number of patents, and Panel B shows the trend for the number of non-self citations per patent. Consistent with the baseline difference-in-differences (DiD) analysis, both measures are normalized by the pre-event averages over the treatment and control firms.

(a) Figure 1: pre- and post-event innovation trends.

Table 3

Difference-in-differences (DiD) test results.

This table reports diagnostics and results of the DiD tests on how exogenous shocks to analyst coverage (brokerage closures and mergers) affect firm innovation activities. The sample begins with all firm-years from 1993 to 2005 with non-missing matching variables and non-missing innovation outcome variables during a seven-year window (from year -3 to year $+3$) around the (actual or matched) event year. For brokerage closures, the treatment firms are those covered by disappearing analysts before the event and whose analyst coverage from these closed brokerages goes down by one. For brokerage mergers, the treatment firms are those covered by both the target and acquirer brokers before the event and for which one of their analysts disappears. The benchmark control portfolios are found by first matching treatment firms to candidate control firms on terciles of market capitalization (MKT), book-to-market (BM), average monthly stock returns (RET), number of analyst following (NOAN), average monthly stock turnover (TURN), and annualized daily stock return volatility (VOL) in the year prior to the event (year -1). For each treatment firm, five controls with the closest analyst coverage in the matching year are then averaged to form a benchmark control portfolio. Panel A reports the univariate comparisons between treatment and control firms' characteristics and their corresponding t -statistics. Panel B gives the full sample DiD test results. Panel C reports the subsample DiD test results based on the number of analysts following the treatment firm in the matching year. Patent is firm's total number of patents in the three-year window before or after the event year. CitePat is firm's total number of future non-self citations received by all patents generated in the three-year window before or after the event year divided by the total number of patents in the corresponding period. To facilitate interpretation, both Patent and CitePat are normalized by pre-event averages over the treatment and the control as described in the text. Standard errors are clustered at the event (deal) level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Post-match differences

Variable	Treatment	Control	Differences	T-statistics
Patent_growth	0.077	0.066	0.011	0.46
CitePat_growth	-0.132	-0.121	-0.011	-0.67
MktCap (MKT), in billions	9.737	10.049	-3.128	-0.65
BookMkt (BM)	0.393	0.388	0.005	0.48
Return (RET)	0.020	0.020	0.000	-0.52
Coverage (NOAN)	15.254	15.299	-0.045	-0.57
Turnover (TURN)	1.289	1.288	0.001	0.03
Annualized Volatility (VOL)	0.492	0.491	0.001	0.24
LnAssets	8.005	7.977	0.028	0.41
LnAge	2.962	2.921	0.041	1.31
ROA	0.137	0.149	-0.012	-1.88*
RETAssets	0.034	0.028	0.006	1.64
InstOwn	0.581	0.569	0.012	1.31
Illiquidity	-5.824	-5.768	-0.056	-1.51

Panel B: Full sample DiD estimators

	Mean treatment difference (after-before) (1)	Mean control difference (after-before) (2)	Mean DiDs (treat-control) (3)
Patent	0.236	0.054	0.182***
(standard error)	(0.042)	(0.033)	(0.053)
CitePat	-0.357	-0.651	0.294***
(standard error)	(0.021)	(0.023)	(0.035)

Panel C: DiD estimators conditioning on initial coverage

	Patent (1)	CitePat (2)
MKT/BM/RET/NOAN/TURN/VOL-matched (coverage≤10)	0.253** (0.106)	0.432*** (0.063)
MKT/BM/RET/NOAN/TURN/VOL-matched (10 < coverage≤25)	0.159** (0.068)	0.269*** (0.047)
MKT/BM/RET/NOAN/TURN/VOL-matched (coverage > 25)	0.112 (0.117)	0.106 (0.093)

(b) Table 3: DiD test results.

6.4 Table 4: Robustness of the DiD results

Read:

- The DiD results remain positive under alternative matching criteria.
- Results are robust to excluding zero-patent firms, changing patent-count thresholds, and changing analyst-coverage thresholds.
- Results remain robust after excluding 2001–2002 events and after using only broker mergers from Hong and Kacperczyk.
- Propensity-score matching also produces positive and significant DiD estimates.

Economic message: The causal result is not an artifact of one matching rule, one subsample, or one innovation threshold.

6.5 Table 5: Instrumental-variable evidence

Read:

- The first-stage coefficient on expected coverage is 0.909 and highly significant.
- The second-stage coefficient on instrumented $LnCoverage$ is -0.115 for patents and -0.137 for citations per patent.
- The IV estimates are larger in magnitude than OLS estimates, suggesting that OLS may be biased upward because innovative firms attract analysts.

Table 4

Robustness tests for Difference-in-differences (DiD) analysis.

This table reports robustness test results for the difference-in-differences analysis. Panel A presents test results using alternative matching criteria. Panel B and C repeat the DiD analysis for subsamples of firms with different thresholds of number of patents and analyst coverage, respectively. Panel D reports test results using alternative subsamples. Panel E adopts a one-to-three propensity score matching method. We first run a probit regression of a dummy variable that equals one if a particular firm-year belongs to our treatment group (and zero otherwise) on a comprehensive list of observable characteristics, including all the independent variables in our baseline regression, stock volatility and turnover, year and firm-French 49 industry dummies, and the growth measures of innovation variables (patents and citations). We then use the predicted probabilities from this probit estimation and perform a nearest-neighbor matching with replacements to form the control group. *Patent* and *CitePat* are defined in Table 3. Standard errors are clustered at the event (*deal*) level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Alternative matching criteria		<i>Patent</i>	<i>CitePat</i>
(1)	Naïve-matched	0.271*** (0.054)	0.477*** (0.045)
(2)	Patent-matched	0.846*** (0.181)	0.768*** (0.076)
(3)	MKT/BM/RET/NOAN-matched	0.286*** (0.060)	0.805*** (0.071)
(4)	MKT/BM/RET/NOAN/TURN/VOL/PAT/CITEPAT-matched	0.595*** (0.172)	0.287** (0.108)
(5)	MKT/NOAN/TURN/VOL-matched	0.293*** (0.042)	0.805*** (0.076)
(6)	MKT/BM/RET/NOAN/TURN/VOL/FF49-matched	0.302*** (0.108)	0.155** (0.057)
(7)	MKT/BM/RET/NOAN/FF49-matched	0.201* (0.110)	0.111** (0.051)
Panel B: Different thresholds for the number of patents			
(1)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>Patent</i> ≥1)	0.331*** (0.078)	0.242*** (0.068)
(2)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>Patent</i> ≥2)	0.327*** (0.063)	0.212*** (0.055)
(3)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>Patent</i> ≥3)	0.345*** (0.057)	0.172*** (0.051)
(4)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>Patent</i> ≥5)	0.343*** (0.065)	0.155*** (0.048)
Panel C: Different thresholds for analyst coverage			
(1)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>coverage</i> ≥5)	0.185*** (0.065)	0.290*** (0.033)
(2)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>coverage</i> ≥15)	0.172** (0.071)	0.227*** (0.049)
(3)	MKT/BM/RET/NOAN/TURN/VOL-matched (<i>coverage</i> ≥25)	0.115 (0.083)	0.105 (0.080)
Panel D: Different subsamples			
(1)	MKT/BM/RET/NOAN/TURN/VOL-matched (excluding 2001–2002)	0.299*** (0.045)	0.549*** (0.083)
(2)	MKT/BM/RET/NOAN/TURN/VOL-matched [Only broker mergers in Hong and Kasperczyk (2010)]	0.159*** (0.046)	0.465** (0.146)
Panel E: Propensity-score matching			
	Propensity score-matched	0.190** (0.076)	0.449*** (0.067)

Table 5

Two-stage least squares (2SLS) regression with the instrument of expected analyst coverage.

This table reports the 2SLS regressions of the innovation outcome variables (three-year-ahead number of patents and number of non-self citations per patent) on analyst coverage, with expected analyst coverage (*ExpCoverage*) as the instrumental variable. The first-stage regression generates the fitted (instrumented) value of *LnCoverage* for use in the second-stage regressions. Definitions of variables are in Panel A of Table 1. Each regression includes year and firm fixed effects. Robust standard errors clustered by firm are displayed in parentheses. The R^2 for the first-stage regressions is a pooled one, and the reported R^2 for the second-stage regressions is a within-firm one. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Dependent variable	First-stage regression	Second-stage regressions	
	<i>LnCoverage</i> (1)	<i>LnPatent</i> _{<i>t</i>+3} (2)	<i>LnCitePat</i> _{<i>t</i>+3} (3)
<i>ExpCoverage</i>	0.909*** (0.012)		
<i>LnCoverage</i> (instrumented)		−0.115*** (0.029)	−0.137*** (0.034)
Controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Number of observations	13,443	13,443	13,443
R^2	0.698	0.147	0.253

Economic message: The IV evidence supports the same conclusion as the DiD design: analyst coverage causally reduces innovation output.

6.6 Table 6: Possible mechanisms

Read:

- Panel A shows that analyst-coverage reductions increase dedicated institutional ownership and reduce transient/quasi-index ownership and takeover exposure.
- Panel B adds mechanism controls to the DiD specification.
- The treatment effect remains positive for both patents and citations.
- This indicates that institutional ownership, takeover exposure, stock liquidity, and discretionary accruals do not fully absorb the analyst-coverage effect.

Economic message: The mechanism evidence supports managerial myopia. Analysts create short-term pressure, and this pressure affects innovation beyond standard governance and market-discipline channels.

Table 6

Possible mechanisms.

This table reports test results on possible mechanisms through which analyst coverage affects innovation, as well as any residual effects of analyst coverage after controlling for direct mechanisms. Panel A presents the difference-in-differences (DiD) results on different categories of institutional ownership and takeover exposure. *Takeover* is the predicted value of *Target* for the following logit regression: $Target_{i,t+1} = \alpha + \beta_1 Q_{i,t} + \beta_2 PPE_{i,t} + \beta_3 \ln Cash_{i,t} + \beta_4 \ln Block_{i,t} + \beta_5 \ln Mkt_{i,t} + \beta_6 \ln DMAs_{i,t} + \beta_7 Leverage_{i,t} + \beta_8 ROA_{i,t} + \beta_9 Year + \beta_{10} Industry + error_{i,t}$, where *Target* equals one if a firm is the target of an attempted or completed acquisition and zero otherwise. All independent variables are constructed in the same way as those in Cremers, Nair, and John (2009). Standard errors are clustered at the event (deal) level. Panel B conducts the DiD analysis on our matched sample (defined in Table 3) in a multivariate framework, after controlling for the dependence of the Di estimator on possible mechanisms. Column 1 regresses *Patent* on a treatment dummy (*Treat*), a dummy to indicate post-event period (*Post*), and the interaction between *Treat* and *Post* (*Treat × Post*). Column 2 regresses *Patent* on *Treat*, *Post*, and *Treat × Post* and controls for *DedOwn*, *TraQixOwn*, *Takeover*, *Illiquidity*, and *DisAccrual*. Column 3 regresses *CitePat* on *Treat*, *Post*, and *Treat × Post*. Column 4 regresses *CitePat* on *Treat*, *Post*, and *Treat × Post* and controls for *DedOwn*, *TraQixOwn*, *Takeover*, *Illiquidity*, and *DisAccrual*. *Patent* and *CitePat* are defined in Table 3. *DisAccrual* is constructed in the same way as in Yu (2008), and we describe the variable construction in Table 1, Panel A. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Changes in different categories of institutional ownership and takeover exposure				
	<i>DedOwn</i>	<i>TraQixOwn</i>	<i>Takeover</i>	
<i>MKT/BM/RET/NOAN/TURN/VOL</i> -matched	0.013*** (0.003)	-0.009** (0.004)	-0.001** (0.000)	
Panel B: DiD tests when controlling for possible mechanism variables				
Dependent variable	<i>Patent</i> (1)	<i>Patent</i> (2)	<i>CitePat</i> (3)	<i>CitePat</i> (4)
<i>Treat × Post</i>	0.204** (0.090)	0.122* (0.072)	0.362*** (0.035)	0.277*** (0.040)
<i>Treat</i>	-0.292*** (0.063)	-0.292*** (0.050)	-0.488*** (0.025)	-0.415*** (0.028)
<i>Post</i>	0.097 (0.061)	-0.017 (0.054)	-0.783*** (0.024)	-0.667*** (0.030)
<i>DedOwn</i>		0.765*** (0.274)		0.614*** (0.154)
<i>TraQixOwn</i>		0.255 (0.160)		0.039 (0.090)
<i>Takeover</i>		-0.107*** (0.013)		-0.036*** (0.007)
<i>Illiquidity</i>		0.063* (0.038)		0.103*** (0.021)
<i>DisAccrual</i>		0.275* (0.160)		0.171* (0.090)
Constant	0.997*** (0.043)	1.516*** (0.113)	1.042*** (0.017)	1.376*** (0.064)
Number of observations	3,863	3,863	3,863	3,863

7 Short summary

- He and Tian ask whether financial analysts encourage or impede corporate innovation.
- The information hypothesis predicts that analysts help innovation by reducing information asymmetry.
- The pressure hypothesis predicts that analysts hurt innovation by increasing short-term earnings pressure.
- Baseline panel regressions show that within-firm increases in analyst coverage reduce future patents and future citations per patent.
- Difference-in-differences tests based on brokerage closures and mergers show that exogenous reductions in analyst coverage increase innovation.
- The DiD effect is strongest among firms with relatively low initial coverage.
- Instrumental-variable estimates using expected coverage confirm a negative causal effect of analyst coverage on innovation.
- Mechanism tests show that analyst-coverage reductions change institutional ownership composition and reduce takeover exposure.
- The treatment effect remains after controlling for these mechanisms, stock illiquidity, and discretionary accruals.
- The main lesson is that analysts can have a dark side: they improve information production but can distort real investment by intensifying managerial myopia.

8 Conclusion

8.1 Core conclusion

He and Tian provide evidence that financial analysts have a negative causal effect on corporate innovation. Firms followed by more analysts produce fewer patents and lower-impact patents, and firms that lose analyst coverage due to plausibly exogenous brokerage events increase innovation.

8.2 Economic intuition

Analysts evaluate firms through near-term forecasts and recommendations. Managers respond to this external pressure by emphasizing short-run earnings and avoiding risky, uncertain, long-horizon projects. Since innovation is precisely such a project, analyst coverage can reduce innovation even while improving information flow to investors.

8.3 Incremental takeaway

The paper's incremental contribution is to show that information intermediaries can shape real investment outcomes. Analyst coverage is not merely a market-efficiency variable; it is also a corporate-governance force. The paper demonstrates that the same financial-market institution that helps investors can impose short-term pressure that weakens technological innovation.